

Fluoride-Rich, Organic-Inorganic Gradient Interphase Enabled by Sacrificial Solvation Shell for Reversible Zinc Metal Battery

Wangwang Xu^{§, a, d}, Jiantao Li^{§, b, *}, Xiaobin Liao^c, Lei Zhang^{§, c}, Xiaoman Zhang^d, Chaozheng Liu^c, Khalil Amine^b, Kangning Zhao^{f, *}, Jun Lu^{g, *}

a College of Materials and Chemical Engineering, China Three Gorges University, Yichang 443002, China.

b Chemical Sciences and Engineering Division, Argonne National Laboratory, Lemont, Illinois 60439, United States.

c State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, International School of Materials Science and Engineering, Wuhan University of Technology, Wuhan 430070, China.

d Department of Mechanical and Industrial Engineering, Louisiana State University, Baton Rouge, LA, 70830, USA.

e Co-Innovation Center of Efficient Processing and Utilization of Forest Resources, College of Materials Science and Engineering, Nanjing Forestry University, Nanjing 210037, China.

f Laboratory of Advanced Separations (LAS), École Polytechnique Fédérale de Lausanne (EPFL), Sion CH-1950, Switzerland.

g College of Chemical and Biological Engineering, Zhejiang University, Hangzhou 310027 (P. R. China).

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ABSTRACT: Zinc metal batteries are strongly hindered by water corrosion since solvated zinc ions would bring the active water molecules to the electrode/electrolyte interface constantly. Herein, we report a sacrificial solvation shell to repel active water molecules from the electrode/electrolyte interface and assist in forming a fluoride-rich, organic-inorganic gradient solid electrolyte interphase (SEI) layer. The simultaneously sacrificial of methanol and $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ would result in the gradient SEI layer with organic-rich surface (CH_2OC - and C_5 product) and inorganic-rich (ZnF_2) bottom, which can combine the merits of fast ion diffusion and high flexibility. As a result, the methanol additive enabled corrosion-free zinc stripping/plating in copper foils for 1000 cycles with an average Coulombic efficiency of 99.95%; a record high cumulative plating capacity of 10 Ah/g at 40 mA/cm² in Zn/Zn symmetrical batteries and, more importantly, at an ultra-low N/P ratio of 2, the practical VO_2 //20- μm -thick-Zn plate full batteries with a high areal capacity of 4.7 mAh/cm² stably operate for over 250 cycles, establishing their promising application for grid-scale energy storage devices. Furthermore, directly utilizing the 20- μm -thick-Zn for the commercial level areal capacity (4.7 mAh/cm²) full zinc battery in our work would simplify the manufacturing process and boost the development of the commercial zinc battery for stationary storage.

INTRODUCTION

Due to their low cost, high ionic conductivity, high safety, and environmental friendliness, aqueous zinc ion batteries (ZIBs) have been extensively studied in recent years, showing great potential to be applied as large-scale energy storage devices.¹ As anode electrode in ZIBs, zinc metal offers a high volumetric capacity of 5855 mAh/cm³, more than 2.5 times the volumetric capacity of lithium metal (2061 mAh/cm³). Coupling non-flammable aqueous electrolytes with zinc metal anode, ZIBs present fast kinetics and high safety.^{2, 3} However, ZIBs do not satisfy the market demands since zinc metal anode suffers from moderate irreversibility during repeated zinc stripping and plating process.^{4, 5} Corrosion of zinc at the interface, mainly due to the active water corrosion⁶ and its related low Coulombic efficiency (CE), are the main drawback to develop zinc anode, especially at high utilization. Moreover, the current research on zinc batteries is usually tested in an ideal state, which cannot represent the actual working conditions for stationary storage.⁷ The balance of cathode and anode in terms of charge and high areal capacity of over 2 mAh/cm² should be taken into account but largely ignored.

Significant research efforts have been dedicated to enhancing zinc electrode reversibility. Numerous approaches have been reported, such as: introducing an electrolyte additive, developing zinc-based alloy, building solid-electrolyte interphase (SEI) film, and using water-in-salt electrolytes. Wang *et al.* developed highly concentrated aqueous electrolytes based on Zn and lithium salts.⁸ By decreasing the number of water molecules in electrolyte, the water-in-salt electrolyte can modulate the Zn²⁺-solvation-sheath structure, leading to significantly improved Zn anode reversibility.^{8, 9} Therefore, tuning the zinc solvation structure would play an important role in adjusting the zinc plating behavior. To tune the zinc solvation structure, electrolyte additive is a promising strategy for practical application due to its low cost and facile preparation process. A tremendous research effort has been dedicated to exploring effective additives. Various electrolyte additives like dimethyl sulfoxide¹⁰, ethylene glycol¹¹, lignin¹², diethyl ether¹³, and so on have also been studied to tune the zinc plating/stripping behavior. Recently, in terms of broadening the working temperature, methanol as an antisolvent additive in aqueous ZnSO₄ electrolyte to realize wide working temperature ZIBs, and they believe that methanol would minimize water activity and weaken Zn²⁺ solvation.¹⁴ They further took a step forward to utilize the triple-function C₃H₇Na₂O₆P to form a self-healing SEI layer to realize the stable cycling of zinc battery¹⁵. Saccharin is introduced to regulate H₂O-poor electrical double layer to create a corrosion-resistant water layer for the zinc anode.¹⁶ More recently, a self-adaptive electric double layer is created to enable high areal capacity (3.38 mA h cm⁻²) cycling of zinc metal battery coupled with a 40-μm-Zn plate¹⁷. Moreover, an alkylammonium salt is doped as an additive in an aqueous electrolyte enabling the in-situ formed solid-electrolyte interphase (SEI) film, which has been demonstrated to increase the electrochemical stability and suppress the H₂ evolution.¹⁸ More recently, methanol has been

directly utilized as solvent for electrolyte with excellent zinc plating/stripping efficiency especially at low temperature of -40 °C in the sacrifice of the high ionic conductivity compared to aqueous electrolyte¹⁹. However, the zinc solvation in an aqueous-based electrolyte will inevitably lead to a direct Zn-H₂O solvation structure and this will bring active water to the electrode/electrolyte interface and in this way, the water corrosion in the zinc plate is regarded as unavoidable.²⁰⁻²¹ Moreover, the design of the SEI layer remains to be unexplored.

Herein, we demonstrate the first report on a sacrificial solvation shell by using methanol as electrolyte additive to enable the formation of a fluoride-rich, organic-inorganic gradient solid electrolyte interface layer and repel the water molecules from the electrode/electrolyte interface for stable zinc plating/stripping. It is found that with the addition of methanol in the aqueous electrolyte, the solvation structure of zinc ions would become Zn(H₂O)₂(CH₃OH)₂²⁺ to bring the methanol molecules rather than water to the electrode/electrolyte interface. The unique sacrificial solvation shell accompanied with the decomposition of Zn(CF₃SO₃)₂ promotes the formation of fluorinate-rich interphase as evidenced by depth profile X-ray photoelectron spectroscopy (XPS) and secondary ion mass spectrometry. In this way, the water corrosion onto the zinc anode is mitigated and the hydrogen evolution rate is suppressed, and a stable interface is formed. As a result, the Zn//Cu asymmetric batteries using Me56 electrolyte deliver an ultra-long cycle life of 1000 cycles (2000 hours) with 99.95% coulombic efficiency, showing excellent reversibility of the Zn plating/stripping process. Even at a high current density of 40 mA/cm² with a capacity of 10 mAh/cm², the Zn//Zn symmetric batteries can deliver a long cycle life of 1000 hours with steady and dendrite-free zinc stripping/plating, corresponding to the ultrahigh cumulative plated capacity (CPC) of 10 Ah/cm². The positive role of methanol additive is further demonstrated in VO₂//Zn full batteries, which deliver a high energy density of 210 Wh/kg at 10 A/g for 3000 cycles and can be operated at various temperatures from -30 to 25 °C. More importantly, at an N/P ratio of 2, the high-areal-capacity VO₂//Zn full batteries (4.7 mAh/cm²) stably operate for over 100 cycles, demonstrating its potential application for grid scale energy storage devices.

RESULTS AND DISCUSSION

The zinc coordination structures in aqueous and M56 electrolyte is studied initially through molecular dynamics (MD) simulation in Figure 1a and 1b. The original Zn(CF₃SO₃)₂ electrolyte shows the bound water close to Zn²⁺ within the average coordination distance of 1.93 Å (Figure 1c), by forming typical Zn(H₂O)₆²⁺. On contrast, after adding methanol, the zinc ions would reduce the coordination number between Zn²⁺ and water through incorporation with CH₃OH molecular by forming Zn(H₂O)₂(CH₃OH)₂²⁺ with the same average coordination distances of 1.93 Å for both Zn-H₂O and Zn-CH₃OH coordination (Figure 1d).

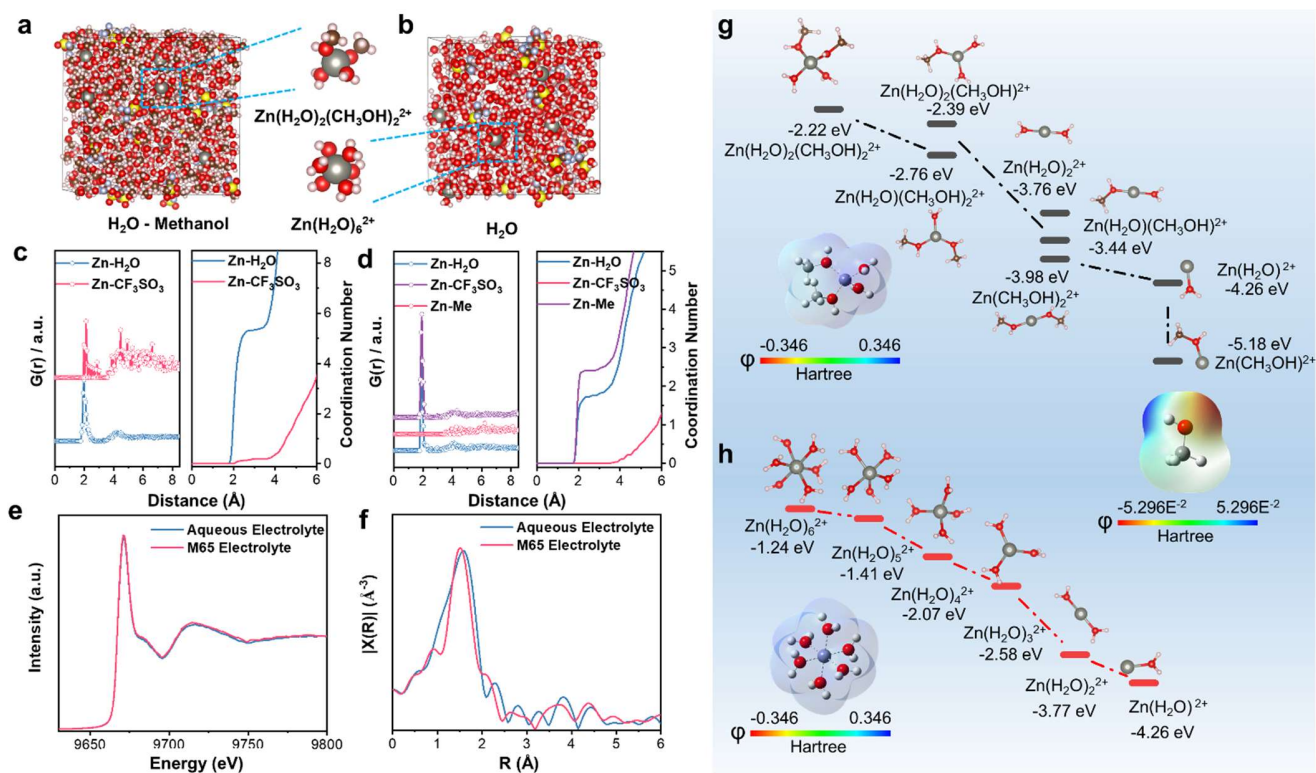


Figure 1. a, b) The MD snapshot of 1 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ in M56 and aqueous electrolyte, respectively. Zn, S, O, C, F, and H atoms are represented by gray, yellow, red, brown, light blue, and pink balls, respectively. c, d) The RDF and the coordination number between Zn^{2+} and H_2O , CF_3SO_3 , and methanol in 1 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ in M56 and aqueous electrolyte, respectively. e, f) The normalized XANES spectra and the corresponding EXAFS spectra of 1 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ in M56 and aqueous electrolyte, respectively. g, h) Calculated binding energies of water and zinc ion and proposed desolvation process of $\text{Zn}(\text{H}_2\text{O})_2(\text{CH}_3\text{OH})_2^{2+}$ and $\text{Zn}(\text{H}_2\text{O})_6^{2+}$.

In order to confirm the solvation shell, the XAFS analysis is carried out to probe the local structure of zinc solvation shell in aqueous and M56 electrolytes. Compared with $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ aqueous electrolyte, M56 electrolyte exhibits a similar X-ray absorption near-edge structure (XANES) spectrum, proving that Zn^{2+} ion shows similar valence state of zinc in the electrolyte (Figure 1e). However, the Fourier transformation of the extended X-ray absorption fine structure (EXAFS) spectrum of M56 electrolyte shows a broad peak at 1.60 Å (Figure 1f), while aqueous electrolyte at 1.72 Å in R space, suggesting the obviously different Zn^{2+} coordination structure (Figure 1h). Additionally, the $\text{Zn}-\text{CF}_3\text{SO}_3$ coordination is very weak, suggesting that CF_3SO_3^- doesn't participate in the Zn^{2+} solvation. We further calculate the sequential binding energies of $\text{Zn}-\text{H}_2\text{O}$ in $\text{Zn}(\text{H}_2\text{O})_2(\text{CH}_3\text{OH})_2^{2+}$ and $\text{Zn}(\text{OH})_6^{2+}$ through DFT to elucidate the desolvation process. By analyzing the binding energy, it can be concluded that the $\text{Zn}(\text{H}_2\text{O})_2(\text{CH}_3\text{OH})_2^{2+}$ would initially lose the two water molecules, while the CH_3OH molecules bind strongly with Zn^{2+} . Such a desolvation process is able to get rid of bringing the water molecules to the electrolyte/electrode interface and in this way, active water-related decomposition in the electrolyte/electrode interface is avoided. On contrast, the desolvation process of $\text{Zn}(\text{H}_2\text{O})_6^{2+}$ would bring the active water molecule to the

electrolyte/electrode interface, which would boost the active water-related decomposition in the electrolyte/electrode interface (Figure 1g). Additionally, the desolvation energy of $\text{Zn}(\text{H}_2\text{O})_2(\text{CH}_3\text{OH})_2^{2+}$ is much lower than that of $\text{Zn}(\text{OH})_6^{2+}$, which is promising for removing the water shell from the solvation shell and repelling water from the electrode/electrolyte interface (Figure 1h).

Wettability on Zn foil directly affects the energy barrier for Zn nucleation behaviors. Therefore, a sessile drop contact angle technique investigated the wettability of different electrolytes. In Figure 2a, with an aqueous electrolyte, Zn metal presents a high contact angle of 73.5°. With the addition of 36 mol % methanol, the contact angle is decreased to 51.2°. With the methanol fraction increased to 56 mol % and 100%, the contact angle is significantly decreased to 29.2° and 19.6°, respectively. These results indicate that methanol has high wettability on zinc surface. Methanol is believed to have a preference to absorb on the surface of zinc metal due to the higher absorption energy, which may significantly impact the Zn nucleation process and the water-induced H_2 evolution.

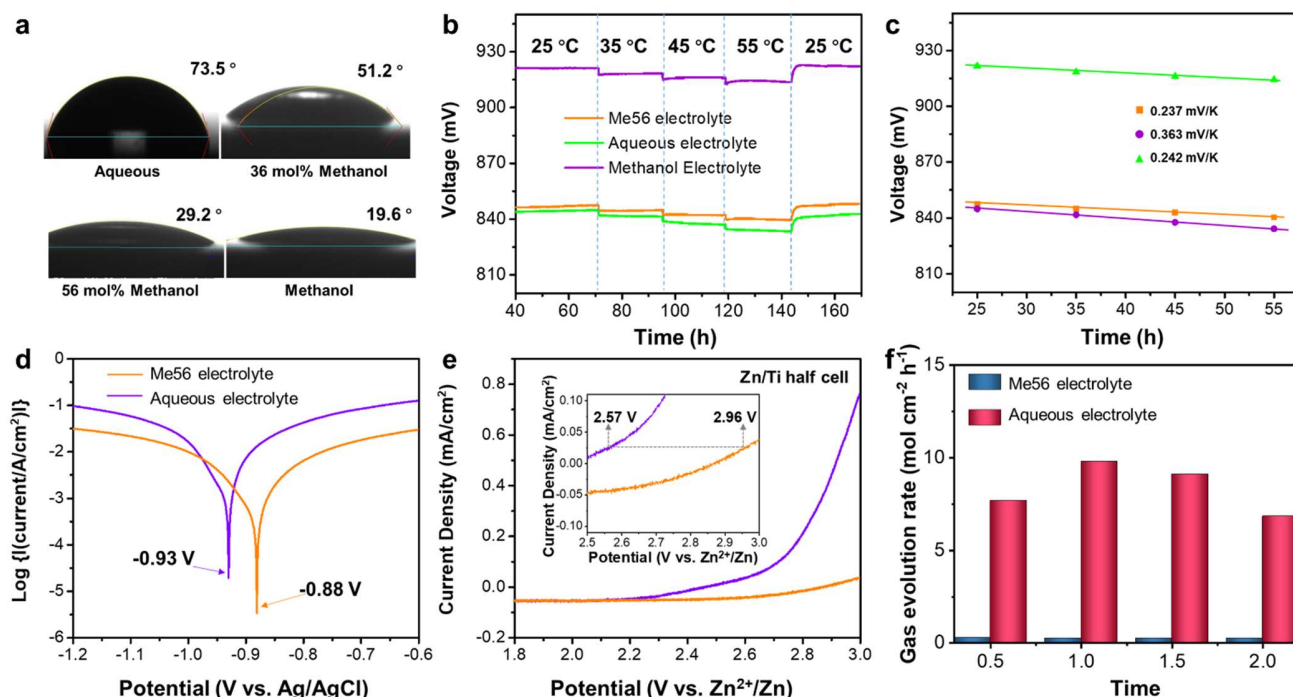


Figure 2. a) Contact angle measurement on zinc metal anode with aqueous electrolyte, Me36 electrolyte, Me56 electrolyte, and pure methanol electrolyte respectively; b) isothermal measurement of VO_2/Zn coin cells under varied temperature; c) corresponding fitting result for the temperature coefficients; d) the Tafel plots of Zn plate tested in Me56 and aqueous electrolyte at a scan rate of 1 mV/s; e) the LSV curves of Zn/Ti half cells tested in Me56 and aqueous electrolyte; f) hydrogen evolution rate quantified by in-situ gas chromatography at a constant current of 10 mA/cm^2 .

Temperature coefficients (TC) were tested under isothermal conditions for the VO_2/Zn coin cells with different electrolytes. Specifically, after cycling at 30 mA/g for 20 cycles, the batteries were charged to 50% state of the total capacity and then placed in an oven. Open circuit voltage (OCV) was tested and recorded for the batteries at different temperatures. As shown in Figure 2b, all the batteries with different electrolytes present a decreased OCV with increased temperature, suggesting negative TCs. TCs of VO_2/Zn full batteries are only related to the entropy change of electrolytes, which present different Zn-ion solvation and desolvation processes. TC results were obtained by fitting the OCV against temperature. As shown in Figure 2c, compared with the TCs value of the batteries with pure aqueous electrolyte (0.363 mV/K), the batteries with Me56 electrolyte and pure methanol electrolyte show the lower TC value of 0.237 mV/K and 0.242 mV/K. This is owing to the smaller entropy change correlated with the addition of methanol into the electrolyte, which is another supportive for the facile desolvation process of $\text{Zn}(\text{H}_2\text{O})_2(\text{CH}_3\text{OH})_2^{2+}$.²² Furthermore, linear sweep voltammetry (LSV) curves are further applied to investigate the corrosion potential of different electrolytes. As shown in Figure 2d, with Me56 electrolyte, Zn plate shows an increased corrosion potential of -0.88 V, which is much higher than that of pure aqueous electrolyte (-0.93 V), confirming the positive role of the methanol additive. Moreover, the oxygen evolution reactions of water significantly confine the upper potential limits of the aqueous electrolyte.

As the LSV curve of Zn/Ti half batteries shown in Figure 2e, the oxygen evolution is suppressed, and the electrochemical window is significantly expanded in comparison with that in aqueous electrolyte. The aqueous electrolyte exhibits an onset potential of 2.21 V for oxygen evolution. With the addition of methanol, the onset potential increases to 2.65 V in M56 electrolyte. In addition, at the current density of 0.025 mA/cm^2 , the cathodic potential was further increased from 2.57 to 2.96 V, showing high stability at high potential. In addition, the potentiostatic current-time transient curve is recorded to study the effects of methanol additive on the nucleation and growth of zinc dendrite. The hydrogen evolution rate is further quantified by in-situ gas chromatography (Figure 2f), and a constant current of 10 mA/cm^2 is applied. The gas evolution in the aqueous electrolyte shows an average hydrogen evolution rate of $\sim 8 \text{ mmol cm}^{-2} \text{h}^{-1}$, while in the Me56 electrolyte, the hydrogen evolution rate is greatly suppressed to $\sim 0.2 \text{ mmol cm}^{-2} \text{h}^{-1}$. The results verify the results from LSV curves, confirming that the corrosion of zinc anode is significantly suppressed. As shown in Figure S1, with a constant overpotential of 150 mV applied, rapidly increased current response can be observed in 20 seconds with a pure aqueous electrolyte.

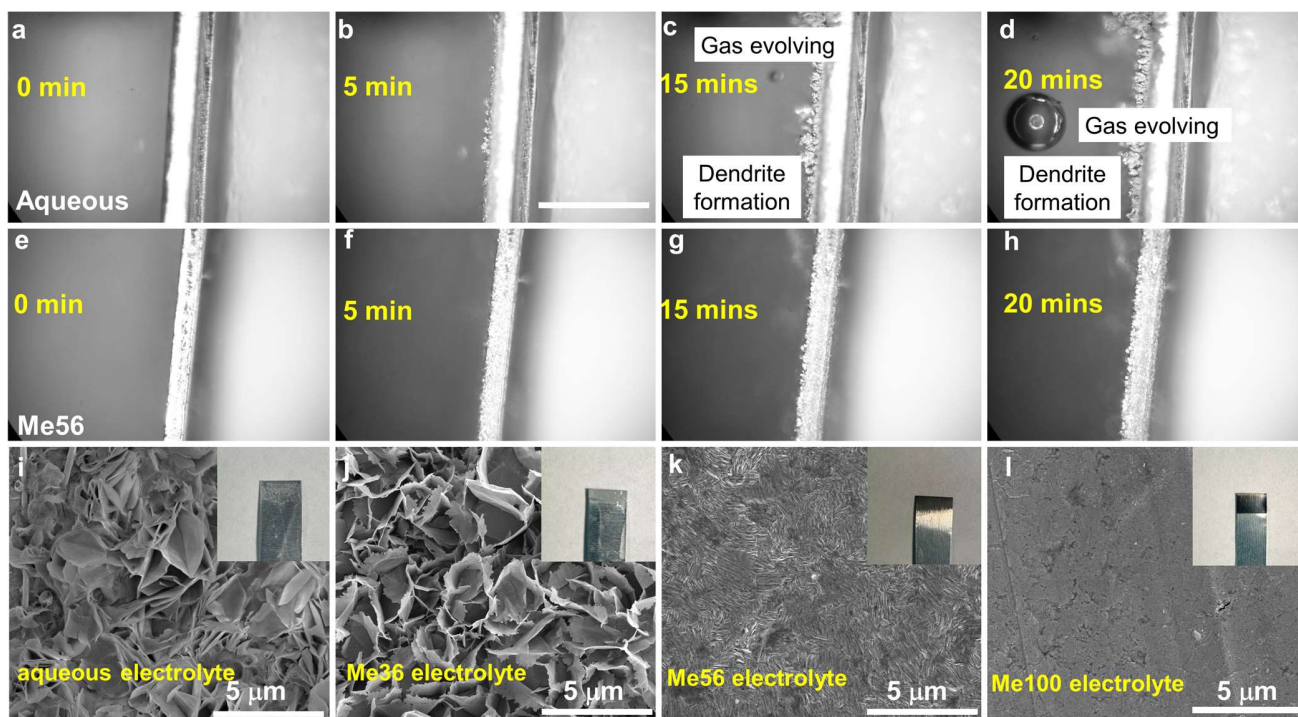


Figure 3. *In situ* optical microscope images of Zn electrode during Zn plating/stripping in a-d) aqueous electrolyte and b-h) Me56 electrolyte. SEM images of planar view of zinc plate after deposition for 1 mAh cm⁻² in i) aqueous electrolyte, j) Me36 electrolyte, k) Me56 electrolyte, and l) Me100 electrolyte, respectively.

With the applied overpotential continuing, the current response increases to more than 30 mA in 300 s. A rise in current density indicates the continuous formation of Zn dendrites, which results in an increased effective surface area. In contrast, for zinc anode cycled in Me56 electrolyte, a stable current density can be observed when overpotential applied. After applying voltage for 300 seconds, the current density merely increased, suggesting dense and smooth zinc deposition.¹⁶ Moreover, the temperature-dependent EIS spectrum of Zn/Zn battery is tested in Figure S2. Notably, the aqueous electrolyte cannot withstand the low temperature of -20 °C, showing the propagating increase in resistance. Furthermore, the Arrhenius plot-based temperature-dependent charge transfer resistance in Figure S3 suggests the low energy barrier of 0.26 eV, suggesting the facile desolvation energy for Me56 electrolyte.

To characterize the deposition behavior, the *in situ* optical microscopic is carried out to *in-situ* observe the zinc plating behavior at a constant discharge current of 1 mA cm⁻² in Figure 3a-3h. The zinc plate in the aqueous solution showed observable dendrite formation after 5-min deposition and large dendrites after 15- and 20-min deposition. Additionally, a large bubble is observed, and it suggests severe gas evolution accompanied, indicating the corrosion of zinc plate, which is consensus with the LSV curves in Figure 2d. On the contrast, the zinc plate in Me56 electrolyte shows minimum dendrite formation, and no bubbles are observed, suggesting uniform zinc deposition and prevention of active water corrosion of zinc plate. In

order to better illustrate the uniform deposition of zinc plate, the SEM image of zinc plate after deposition at 1 mAh cm⁻² in different electrolytes (aqueous electrolyte, Me36, Me56, and Me100 electrolyte) is shown in Figure 3i-3l. With the increasing amount of methanol, the surface of zinc plate becomes more and more smooth. In the aqueous electrolyte (Figure 3i), the surface is covered with plates which is vertical to the zinc plate. In contrast, in Me56 electrolyte, the surface is pretty smooth. Additionally, we notice that in Me56 electrolyte, the zinc plate becomes black, while no color change is observed in aqueous electrolyte, suggesting the possible solid electrolyte interface (SEI) formation in methanol-containing electrolyte. To probe the possible formation of SEI layer, X-ray photoelectron spectroscopy (XPS) is carried out to analyze the composition. In the C1s spectrum (Figure 4a), the surface is consisted of -CF₃ (288.7 eV), which originated from electrolyte residuals of anion CH₃SO₃⁻, while with the increased time of Ar⁺ etching, the -CF₃ and C=C peaks disappeared and the -CO₃ peaks at 290.3 eV increases, suggesting the formation of ZnCO₃, which is believed to be beneficial for reversible zinc deposition.^{18, 23-24} This is also confirmed by the O 1s spectrum (Figure 4b), where carbonate and hydroxide peak become dominate. Additionally, the reduced peaks for both C-C and C-O suggest methanol decomposition.

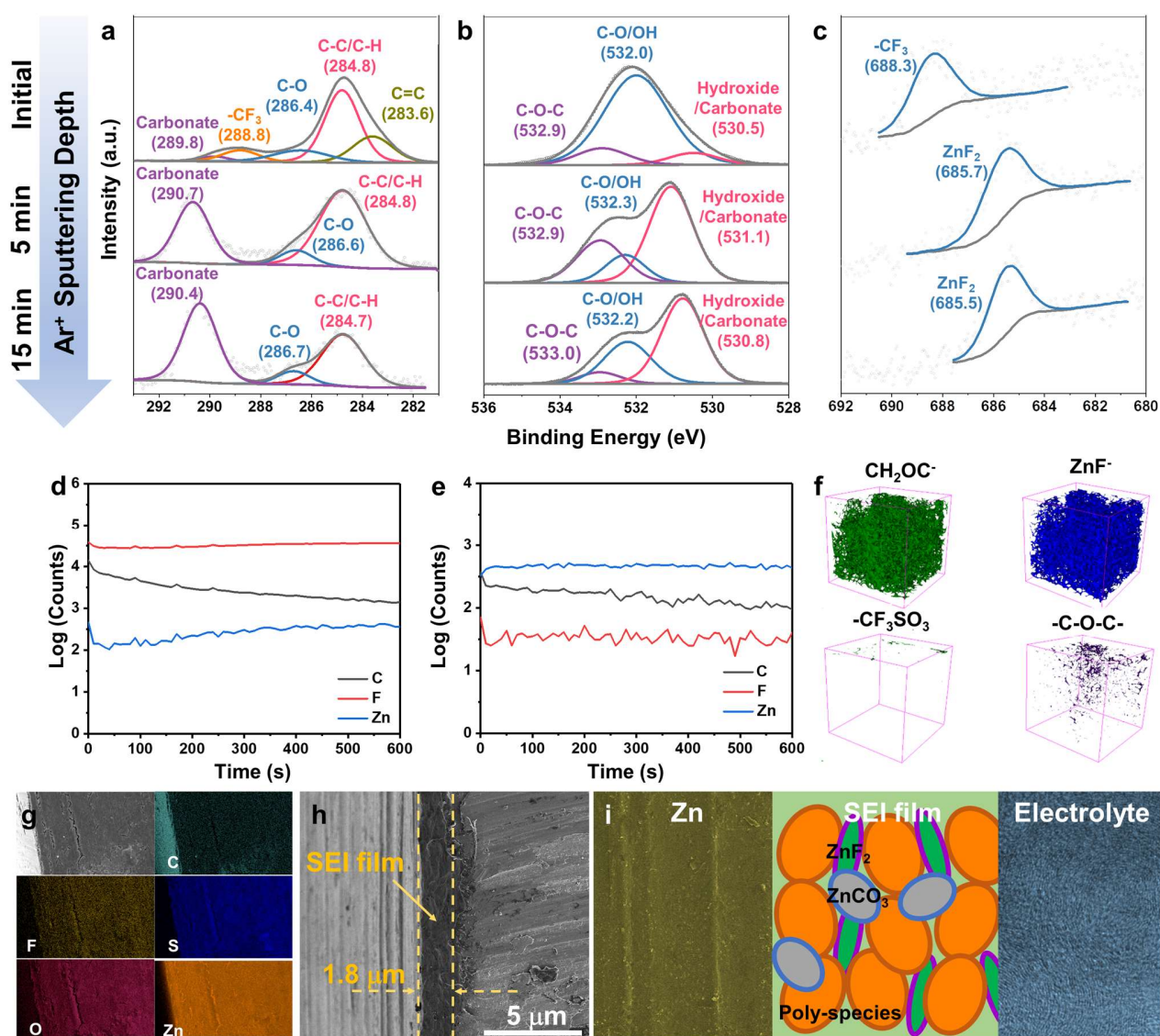


Figure 4. Depth profiles of a) C1s, b) O1s, and c) F1s spectrum generated after Ar^+ sputtering for 0 min, 5 min, and 15 min. d) Proposed mechanism demonstrating synergistic reactions of methanol and CF_3SO_3^- to deposit predominantly fluoride and carbonate-based SEI. Depth profile curves of TOF-SIMS in d) negative mode and e) positive mode. f) 3D variation of the TOF-SIMS intensity related to the corresponding charged fragments. g) EDX mapping of the SEI layer. h) Cross sectional SEM image of Zn plate after 10 cycles. i) Proposed schematic of Zn^{2+} -conducting SEI which is composed of small nodular particles embedded in a polymeric framework.

Moreover, the F1s spectrum (Figure 4c) suggests the underlying ZnF_2 (284.9 eV) after Ar^+ bombardment, suggesting a ZnF_2 -rich SEI layer, which is reported to be able to suppress dendrite formation from growing in the vertical direction due to the high interface energy and prevent the surface from active water corrosion and its related parasitic reaction. We proposed the ZnF_2 -rich SEI layer reaction route based on the pathway from dimethyl ether in the weak acid electrolyte.²⁵ The triflate would be reduced through the breaking results above.

Initially, methanol will undergo the dissociative of C-F bonds and generate $-(\text{SO}_3)\text{F}_2\text{C}$ species, which reacts with dimethyl ether through highly exergonic polymerization reaction by forming large polyanion and ZnF_2 .²⁵ The formation of ZnCO_3 is believed to be related to the dissolved CO_2 in the electrolyte.

This is also confirmed by the EDX mapping of the SEI layer, which contains F, S, C, and O in the surface (Figure 4e). The cross-sectional SEM image reveals the uniform SEI film with a thickness of 1.8 μm (Figure 4f). In summary, the SEI layer comprises polyanion species along with ZnCO_3 and ZnF_2 , precipitating on zinc electrode/electrolyte interface *in situ*, which is believed to provide physical protection against active water corrosion²⁶ (Figure 4g).

Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS) using a Ga^+ ion beam was further carried out to characterize the SEI layer. The depth profile of the SEI layer in the negative and positive scan mode shows the fluoride rich nature and the fluoride percentage remains to be quite high after 600 s etching with no obvious decay in Figure 4d and 4e, respectively.

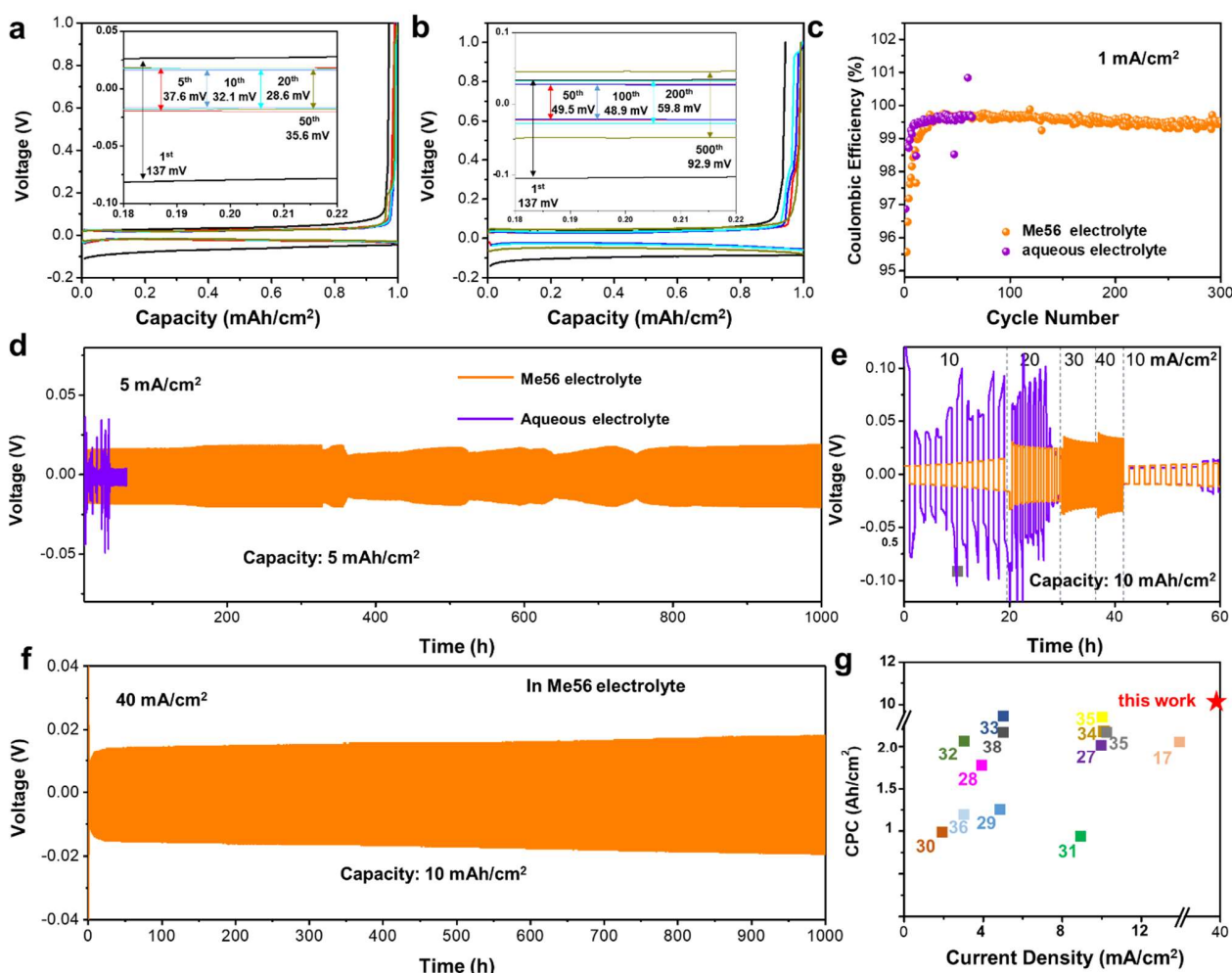


Figure 5. Zn plating/stripping profiles on copper foil in a) aqueous electrolyte and b) Me56 electrolyte at 1 mA/cm². (inset with magnified views of selected cycles between 0.18 and 0.22 V). c) Coulombic efficiencies of Zn/Cu half cells with different electrolytes at 1 mA/cm² current density. d) Long-term galvanostatic Zn plating/stripping in Zn//Zn symmetric batteries at 5 mA/cm² with specific capacity of 5 mAh/cm². e) The rate performance of Zn//Zn symmetric batteries at the current density of 10, 20, 30, and 40 mA/cm² with the capacity of 10 mAh/cm². f) Long-term galvanostatic Zn plating/stripping in Zn//Zn symmetric batteries in Me56 electrolyte at a 40 mA/cm² with capacity of 10 mAh/cm². g) The CPC comparison of Zn symmetric cell using Me56 electrolyte with other reported literatures.

The organic contribution as a sign of carbon gradually reduced while the zinc amount slightly increases with the etching, suggesting the gradient nature of the SEI layer with surface rich organic component while inside inorganic rich inner. We further plotted the possible components including the organic staff (CH₂OC-, -CF₃SO₃, and -C-O-C) and inorganic part (ZnF⁻). The major components are CH₂OC- as organic SEI layer and ZnF⁻ as the inorganic part. The possible intermediate -C-O-C- is only observed in the surface part, suggesting the possible highly ex-ergonic polymerization of methanol in the electrolyte.

We further quantify the polymerization product by comparing the depth profiles of C₁, C₂, C₃, C₄, C₅, and C₆ (Figure S4). The C₅ product is the major product without decreasing with the depth etching. The 3D images in Figure S5 confirms the major composition of the C₅ product in the area, confirming the polymerization of methanol as a role for sacrificial solvation shell. Additionally, the -CF₃SO₃ is only observed in the surface, suggesting that reaction might be initiated by the decomposition

of -CF₃SO₃. All these results above confirm that the SEI layer in the interface is the decomposition of -CF₃SO₃, and the polymerization of methanol. This reaction route leads to the fluoride-rich, organic-inorganic gradient SEI layer, which is beneficial for the reversible cycling of the zinc metal battery.

The reversibility of zinc metal anode in the different electrolytes was investigated in Zn/Cu half cells. As shown in Figures 5a and 5b, the addition of methanol slightly increases the overpotential of Zn stripping and plating on the copper foil surface. This is attributed to the improved interfacial resistance and reduced ionic conductivity at room temperature. However, with Me56 electrolyte, the Zn//Cu batteries demonstrate a high average Coulombic Efficiency (CE) up to 98% for 300 cycles at the current of 1 mA/cm² in Figure 5c. By contrast, Zn plating/stripping in 1 M aqueous electrolyte presents a low CE of 80%, while the battery fails at 57th cycle.

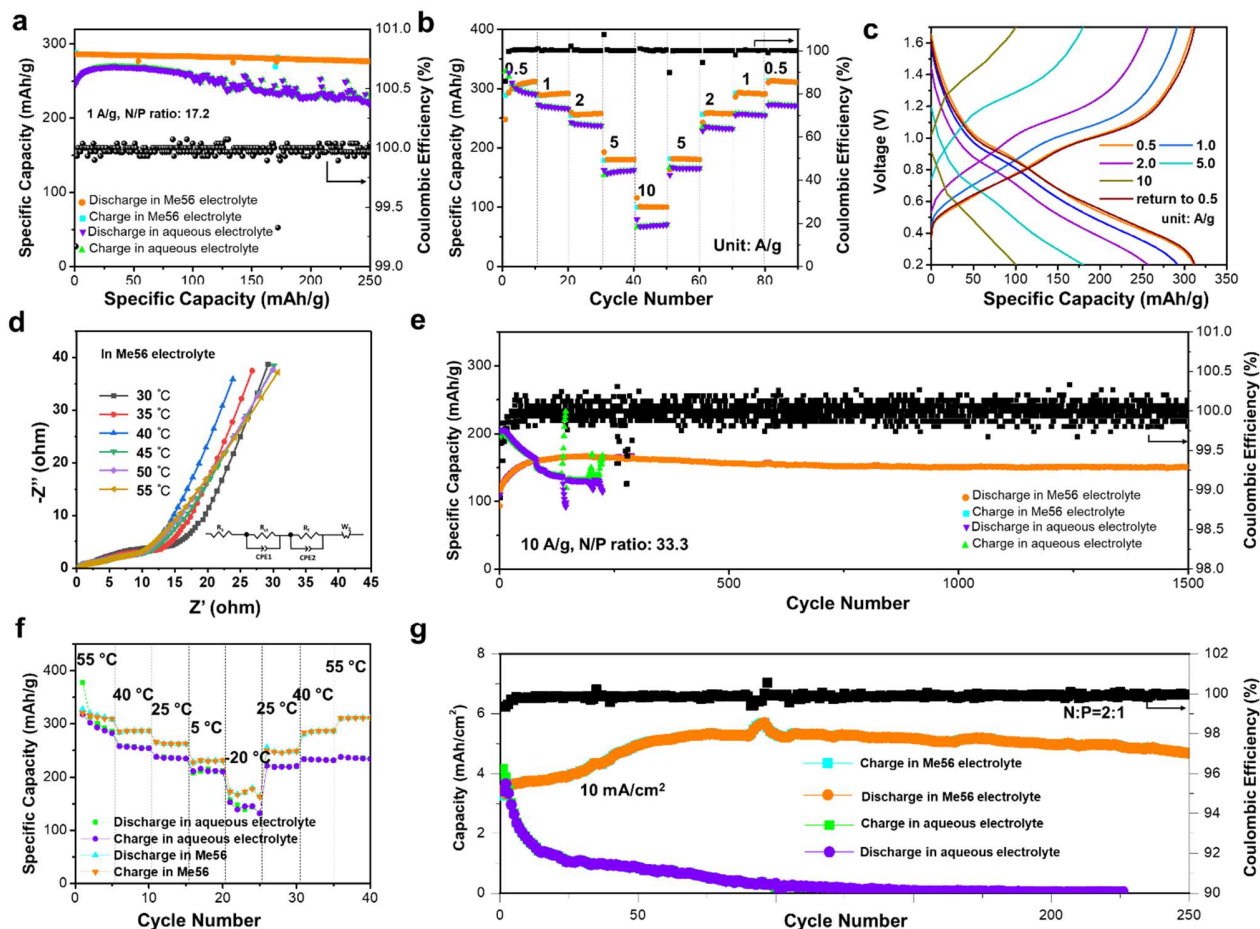


Figure 6. Electrochemical performance of VO₂/Zn batteries with Me56 and aqueous electrolytes. a) Cycling performance at 1 A/g. b) Rate performance with the current density ranging from 0.5 to 10 A/g. c) Charge-discharge profiles at different current density. d) EIS spectra of the battery after 5 cycles at different temperature in M56 electrolyte. e) Long cycling performance at the high current density of 10 A/g. f) Cycling stability of VO₂/Zn batteries at different temperature ranging from -20 to 55 °C. g) Practical VO₂/Zn battery at N/P ratio of 2 and high current of 10 mA/cm² at high areal capacity of 4.7 mAh/cm².

When the areal capacity is increased to 5 mAh/cm² and the current is increased to 5 mA/cm², the CE for Zn plating and stripping in Me56 electrolyte is still maintained at > 99% for over 100 cycles (Figure S6). This is also confirmed by an even higher current (10 mA/cm²) and higher areal capacity (10 mAh/cm²) (Figure S7). The addition of methanol exhibits excellent Zn²⁺ reversibility at a high current in both Cu//Zn half cells and Zn//Zn symmetric cells. At the current of 5 mA/cm² with an areal capacity of 5 mAh/cm², Me56 electrolyte rendered excellent Zn²⁺ reversibility in Zn//Zn symmetric cells for over 1000 hours (Figure 5d).

Moreover, the rate performance of symmetric Zn cells with/without methanol additive is also studied at the high current density ranging from 10 to 40 mA/cm² with the areal capacity of 10 mAh/cm² (Figure 5e). Clearly, Zn//Zn symmetric cells with Me56 electrolyte show a flat and steady voltage plateau with the overpotential of about 12 mV at each current density. With the current density increased from 10 to 40 mA/cm², the cells with Me56 electrolyte maintain steady voltage profiles at each current stage, and the overpotential increased to 35 mV at 40 mA/cm². However, the cells using aqueous electrolytes fail after 30 cycles at 30 mA/cm². When the current density returns to 10 mA/cm², the cells with Me56 electrolyte still work

and return to the initial steady voltage, but the cells with aqueous electrolyte fail immediately. The cycling stability at super-high current density is further studied. The high reversibility is achieved at a low current of 0.5 mA/cm² with 0.25 mAh/cm² with the presence of a methanol additive (Figure S8). Furthermore, in Figure 2i, with Me56 electrolyte, the Zn symmetric cells can achieve a long cyclic lifespan of 1000 hours at a high current density of 40 mA/cm² with an areal capacity of 10 mAh/cm², revealing the high reversibility of Zn plating and stripping at high current. By contrast, the Zn symmetric cells with pure aqueous electrolyte short-circuited after only two cycles (Figure S9). The cumulative plated capacity (CPC) value is further proposed to accurately evaluate the stability of symmetric cells.¹⁶ Figure 5g presents the CPC value of the recently reported literatures and the results of our work. Encouragingly, with Me56 electrolyte, the CPC achieves an ultrahigh value of 10 A/cm² at the current density of 40 mA/cm², surpassing all other reported results^{17, 27-38}.

To further investigate the efficiency of the methanol additive, VO₂/Zn full batteries were assembled with Me56 and aqueous electrolytes, respectively. As shown in Figure S10a, all the CV curves at different scan rates show two pairs of redox peaks, which are attributed to the insertion/extraction of zinc

ions into/out of VO_2 lattice. For both Me56 and aqueous electrolytes, the peak position and the shape of CV curves have few differences, revealing the same electrochemical reactions with different electrolytes. Figure 6a shows the cycling performance at the current density of 1 A/g. After 100 cycles, the batteries with Me56 electrolyte deliver the specific capacity of 283.4 mAh/g, maintaining 98% of the initial specific capacity (288.8 mAh/g), exhibiting high capacity and favorable cycling stability. The batteries with pure aqueous electrolyte provide a capacity of 183.1 mAh/g after 100 cycles, retaining only 57.7% of the initial capacity (317 mAh/g). Figure 6b shows the rate performance of batteries, and Figure 6c illustrates the corresponding galvanostatic charge-discharge curves. At the current densities of 0.5, 1, 2, 5, and 10 A/g, batteries with Me56 electrolyte deliver the specific capacity of 311, 290, 256, 180, and 100 mAh/g, respectively, showing much higher specific capacity than those batteries with aqueous electrolyte at each current density. Thus, to further quantify the resistance of SEI and charge transfer, the EIS spectra after 5 cycles in different electrolytes are presented in Figure 6d and S9. To share in-depth insight in the kinetics of the SEI and charge transfer, temperature dependent experiment is carried out from 30 to 55 °C. All the EIS spectra show two semicircles and one inclined curve. The first semicircle at high frequency represents the SEI related resistance while the second at rather lower frequency represents the charge transfer resistance (e. g. desolvation process, zinc ion transport in the electrolyte). All the semicircles in the M56 electrolyte are smaller than those in the aqueous electrolyte. This suggests that despite the lower zinc ion conductivity in the M56 electrolyte, the charge transfer resistance is still lower, which indicate that the desolvation process is the rating limiting step for charge transfer resistance rather than zinc ion conductivity. The reduced resistance is attributed to the promoted interfacial charge transfer kinetics with reorganized Zn^{2+} solvation sheath, enabling stable and reversible cycling and fast (de)intercalation of Zn^{2+} into cathodes. Furthermore, the temperature dependent EIS suggested that the SEI and charge transfer resistances follows temperature activated process. Thus, we are able to apply Arrhenius equation to calculate the activation energy of the SEI and charge transfer resistances. Both activation energy (7.2 kJ/mol, and 11.2 kJ/mol) in M56 are substantially lower than their counterparts in aqueous electrolyte, further confirming the advantage of the methanol assisted fluoride-rich, organic-inorganic gradient solid electrolyte interface layer as well as facile methanol incorporated desolvation process.

Therefore, the ultralong cycling stability is achieved at the high current density of 10 A/g. As displayed in Figure 6e, with Me56 hybrid electrolyte, the VO_2/Zn batteries deliver the high specific capacity of 152 mAh/g after 1500 cycles with almost 100% Coulombic efficiency (CE), retaining about 90% of the highest capacity (180th cycle). However, with aqueous electrolyte, the capacity decays a lot in the first 100 cycles, and finally, the cell fails at about 200th cycle. The intercalation behavior of zinc ions in the cathode electrode is further studied. As illustrated in Figure S10, with the scan rate increased from 0.1 to 2.0 mV/s, the anodic peaks move to a higher voltage and the cathodic peaks shift to a lower voltage. This is owing to the increased diffusion resistance at a higher scan rate. The electrochemical kinetics were analyzed *via* the power law formula in equation S1. As observed in Figure S10b, the *b* value from the slope of peak 1 and 2 are 0.899 and 0.785, implying that both diffusion and capacitive processes contribute to the capacity.

The contribution ratios at different scan rates are further calculated based on equations S2 and S3. As displayed Figure S10b, at the scan rate of 0.1 mV/s, ~51% of the capacity is contributed by the capacitive process, suggesting that the electrochemical reactions are main affected by the capacitive process. In order to probe the high utilization of zinc anode, cycling performance with low N/P ratio of 2 were tested at a high areal capacity of 4.7 mAh/cm². The assembled cell with Me56 electrolyte shows excellent cycling stability for 250 cycles at 10 mA/cm², maintaining a Coulombic Efficiency of 100% (Figure 6g). The initial capacity increase is originated from the thick electrode and high current density. In contrast, the capacity of the cell in aqueous electrolyte decays quickly in the initial 20 cycles, and the capacity becomes minimum afterward. The $\text{VO}_2/10\text{-}\mu\text{m}$ -thick Zn cell with an ultralow N/P ratio of 1 is also tested as a proof-of-concept, and the cell is able to operate for 100 cycles with 1.1 mAh/cm² retained (Figure S11). To investigate the practicability of electrolyte additive, pouch cells were assembled with different electrolytes and tested their cycling performance with low N/P ratio of 2.5 at a high areal capacity of 2.2 mAh/cm². The assembled pouch cells with Me56 electrolyte show excellent cycling stability for over 100 cycles at 1 A/g, retaining about 82% of the initial capacity, and maintaining a Coulombic Efficiency of approaching 100% (Figure S13). In contrast, the capacity of pouch cells with aqueous electrolyte plummeted to a mere 155 mAh/g after 100 cycles. Notably, a rapid decline in capacity was observed as early as the 53rd cycle, indicating the depletion of the zinc anode. This finding further solidifies the exceptional reversibility of zinc in the presence of Me56 electrolyte.

CONCLUSION

In summary, we have successfully demonstrated the concept of a sacrificial solvation shell to retard the active water corrosion and facilitate the fluoride-rich, organic-inorganic gradient SEI layer to enable highly reversible zinc metal battery. The methanol would promote the formation of $\text{Zn}(\text{H}_2\text{O})_2(\text{CH}_3\text{OH})_2^{2+}$ solvation structure. During the sacrificial desolvation process, two methanol molecules would bind strongly with zinc ions, leading to the $\text{Zn}(\text{CH}_3\text{OH})_2^{2+}$ structure, preventing the solvated water molecules from the interface. The simultaneously sacrificial of methanol and $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ would result in the gradient SEI layer with organic-rich surface ($\text{CH}_2\text{OC}-$) and inorganic-rich bottom, which can combine the merits of strong fast diffusion and high flexibility. As a result, the sacrificial solvation shell enabled dendrite-free zinc stripping/plating in copper foil for 1000 cycles with coulombic efficiency of 99.95%; ultrahigh cumulative plating capacity of 10 Ah/g at 40 mA/cm² in Zn/Zn symmetrical battery. More importantly, at the low NP ratio of 2, the high-areal-capacity $\text{VO}_2/20\text{-}\mu\text{m}$ -Zn plate full cell with ultra-high areal capacity of 4.7 mAh/cm² shows stable cyclability for over a hundred cycles, establishing a viable option for practical application. Our work paves a significant step towards commercialized zinc metal batteries by direct utilizing a 20- μm -Zn plate to achieve commercial-level areal capacity, which would simplify the manufacturing process and boost the development of commercialized zinc batteries.

ASSOCIATED CONTENT

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Detailed experimental procedures, characterizations, computational methods, supporting figures including potentiostatic current-time transient curves, charge transfer resistance and calculation of activation energy, coulombic efficiencies of Zn/Cu half cells, cycling performance of Zn//Zn symmetric batteries and VO₂/Zn pouch cells, CV curves, depth profile curves and 3D images of TOF-SIMS analysis.

AUTHOR INFORMATION

Corresponding Author

*Kangning Zhao - Laboratory of Advanced Separations (LAS), École Polytechnique Fédérale de Lausanne (EPFL), Sion CH-1950, Switzerland. Email: kangning.zhao@epfl.ch; vicvel@126.com

*Jiantao Li - Chemical Sciences and Engineering Division, Argonne National Laboratory, Lemont, Illinois 60439, United States. Email: jiantao.li@anl.gov; jiantao_work@126.com

*Jun Lu - College of Chemical and Biological Engineering, Zhejiang University, Hangzhou 310027 (P. R. China). Email: junzoelu@zju.edu.cn

Author Contributions

\$W.X. and J.L. contributed equally.

Notes

The authors declare no competing financial interest.

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